

Second Question bank ETI

Q1. What is IoT?

Ans: IoT means Internet of Things — it connects devices to share data.

Q2. Define IoT.

Ans: IoT is a network of interconnected devices that collect and exchange data over the internet.

Q3. State any three characteristics of IoT.

Ans:

1. **Connectivity** – Devices are connected to the internet.
2. **Automation** – Devices work with little human help.
3. **Data Sharing** – Devices share and collect useful data.

Q4. Explain the definition and characteristics of IoT.

Ans:

IoT stands for Internet of Things. It is a system where physical objects like sensors, devices, and machines are connected to the internet. These devices collect and exchange data to perform tasks smartly. Its main characteristics are connectivity, automation, data collection, real-time monitoring, and remote control of devices.

Q5. What are IoT enabling technologies?

Ans: Technologies that help build and run IoT systems are called IoT enabling technologies.

Q6. Name any two IoT enabling technologies.

Ans:

1. Wireless Sensor Networks (WSN)
2. Cloud Computing

Q7. Mention any three IoT enabling technologies with functions.

Ans:

1. **Sensor Technology** – Detects physical changes like temperature or motion.
2. **Cloud Computing** – Stores and processes data remotely.
3. **Internet Connectivity** – Enables communication between devices.

Q8. Explain any five IoT enabling technologies.

Ans:

1. **Sensors** – Collect real-world data like temperature, light, etc.
2. **Internet** – Helps devices connect and communicate.
3. **Cloud Computing** – Stores and analyzes large amounts of data.
4. **Big Data** – Manages huge data from devices.
5. **Artificial Intelligence (AI)** – Makes smart decisions using data.

Q9. What is meant by IoT levels?

Ans: IoT levels show different stages in building an IoT system.

Q10. What is an IoT deployment template?

Ans: It is a basic structure or model used to set up an IoT system.

Q11. Name two IoT levels.

Ans:

1. Perception Level
2. Network Level

Q12. Name any two types of deployment templates in IoT.

Ans:

1. Smart Home Template
2. Smart City Template

Q13. Mention three levels of IoT and their functions.

Ans:

1. **Perception Level** – Collects data using sensors.
2. **Network Level** – Transfers data to servers.
3. **Application Level** – Uses data to provide services.

Q14. Explain different levels of IoT and mention any two deployment templates.

Ans:

IoT has three main levels:

1. **Perception Level** – Gathers data through sensors.
2. **Network Level** – Sends data through the internet or other networks.
3. **Application Level** – Uses data to perform smart tasks for users.

Two deployment templates are:

- **Smart Home Template** – Used in home automation systems.
- **Smart City Template** – Used for traffic, waste, and public safety management.

Q15. What is one major challenge in IoT?

Ans: Security is a major challenge in IoT.

Q16. Give one application of IoT.

Ans: Smart Home is one application of IoT.

Q17. Name two challenges of IoT.

Ans:

1. Data security
2. Internet connectivity issues

Q18. Name any one IoT device.

Ans: Arduino Uno

Q19. What is NodeMCU used for?

Ans: It connects devices to Wi-Fi in IoT projects.

Q20. Name any two IoT devices.

Ans:

1. Arduino Uno
2. Raspberry Pi

Q21. What is Arduino Uno?

Ans:

Arduino Uno is a microcontroller board used to control sensors and devices.

Q22. Write features of any two IoT devices.

Ans:

Arduino Uno: Easy to program, controls sensors, has digital pins.

Raspberry Pi: Acts like a mini computer runs Linux, has USB and HDMI ports.

Q23. Compare Arduino Uno, Raspberry Pi, and NodeMCU with features.

Ans:

1. **Arduino Uno** – Microcontroller board, simple to use, controls sensors.
 2. **Raspberry Pi** – Minicomputer supports internet, good for complex tasks.
 3. **NodeMCU** – Has built-in Wi-Fi, great for wireless IoT, low power usage.
- Each device has its own use depending on the project type and complexity.

Q24. Name any two applications of IoT.

Ans:

1. Smart Healthcare
2. Smart Agriculture

Q25. Write any three issues or challenges in IoT.

Ans:

1. **Security** – Protecting data from hacking.
2. **Privacy** – Keeping user data safe.
3. **Power Usage** – Devices need constant energy supply.

Q26. Explain any five IoT issues and challenges.

Ans:

1. **Security** – Devices are vulnerable to cyber-attacks.
2. **Privacy** – User information can be misused.
3. **Data Management** – Handling large data is difficult.
4. **Connectivity** – Poor internet affects performance.
5. **Power Consumption** – Devices need long-lasting power sources.

Q27. Explain any five applications of IoT.

Ans:

1. **Smart Home** – Controls lights, fans, and security remotely.
2. **Smart Healthcare** – Monitors patient health in real time.
3. **Smart Agriculture** – Tracks crop health and soil moisture.
4. **Smart City** – Manages traffic, waste, and lighting.
5. **Industrial IoT** – Monitors machines and reduces downtime.

Q28. What is Blockchain?

Ans: Blockchain is a digital ledger that stores data in blocks linked in a chain.

Q29. Why is Blockchain secure?

Ans: Blockchain is secure because data is stored in blocks that cannot be changed easily.

Q30. Mention any three features of Blockchain.

Ans:

1. **Decentralized** – No single owner controls it.
2. **Secure** – Data is locked and hard to hack.
3. **Transparent** – Everyone can view the records.

Q31. Explain Blockchain and its important features.

Ans:

Blockchain is a secure digital record system where data is saved in blocks connected in a chain. It is decentralized, meaning no single person controls it. It ensures security, transparency, and data cannot be changed once added. It is mainly used in cryptocurrencies

Q32. Who invented Blockchain?

Ans: Satoshi Nakamoto

Q33. When Blockchain was first introduced?

Ans: Blockchain was introduced in 2008 with Bitcoin.

Q34. Why was Blockchain created?

Ans:

Blockchain was created to support Bitcoin and allow secure, transparent, and decentralized digital transactions without a middleman.

Q35. Trace the origin of Blockchain technology.

Ans:

Blockchain was first introduced in 2008 by a person or group using the name **Satoshi Nakamoto**. It was created to support **Bitcoin**, a digital currency. The aim was to create a **decentralized and secure** system for recording transactions. Over time, Blockchain became popular beyond Bitcoin and is now used in banking, supply chains, healthcare, and more.

Q36. What is one problem with current transaction systems?

Ans: High transaction fees

Q37. Name two shortcomings of traditional transaction systems.

Ans:

1. Slow processing time
2. Risk of fraud

Q38. Mention any three drawbacks of current transaction systems.

Ans:

1. **Centralized control** – Managed by banks or third parties.
2. **Limited transparency** – Users can't fully track transactions.
3. **Security risks** – Prone to hacking and fraud.

Q39. Explain the shortcomings of current transaction systems.

Ans:

Current transaction systems have several issues. They are **centralized**, meaning controlled by banks or third parties, which may lead to **delays** and **extra charges**. They also lack full **transparency**, making it hard to track issues. Additionally, they are at **risk of fraud**, **hacking**, and often involve **high transaction fees** and **slow processing times**, especially for international payments.

Q40. What is Bitcoin?

Ans: Bitcoin is a digital currency created using blockchain technology.

Q41. Who created Bitcoin and when?

Ans:

Bitcoin was created by **Satoshi Nakamoto** in **2008**.

Q42. Why was Bitcoin introduced?

Ans:

Bitcoin was introduced to allow **peer-to-peer digital payments** without involving banks. It aimed to solve issues like high fees, fraud, and delays in traditional transactions.

Q43. Explain the emergence of Bitcoin.

Ans:

Bitcoin was introduced in **2008** by an unknown person or group named **Satoshi Nakamoto**. It was created to provide a **decentralized digital currency** that works without banks. Bitcoin uses **blockchain** to record transactions securely. It solved problems in traditional systems like **high fees**, **slow transfers**, and **fraud**. It was the first real-world use of blockchain technology.

Q44. What is a cryptographic key?

Ans: It is a code used to secure and access blockchain data.

Q45. Name the two types of cryptographic keys used in blockchain.

Ans:

1. Public Key
2. Private Key

Q46. Differentiate between public key and private key.

Ans:

A **public key** is shared openly and used to receive data. A **private key** is secret and used to access or sign data. Both work together to ensure secure transactions.

Q47. Explain cryptographic keys in blockchain and their use.

Ans:

Blockchain uses **public and private keys** for secure transactions. A **public key** is like an address that others can see to send you data. A **private key** is kept secret and used to unlock or sign that data. Together, they ensure **authentication, privacy, and data security** in blockchain systems. Without the private key, no one can access the stored data.

Q48. What is a network protocol in blockchain?

Ans: A network protocol in blockchain defines the rules for communication between nodes.

Q49. Name any two network protocols used in blockchain.

Ans:

1. **Bitcoin Protocol**
2. **Ethereum Protocol**

Q50. What is the role of network protocols in blockchain?

Ans:

Network protocols in blockchain ensure secure, standardized communication and data transfer between nodes, enabling decentralized transactions and consensus.

Q51. Explain the role of network protocols in blockchain.

Ans:

Network protocols in blockchain define the rules and procedures for communication between blockchain nodes. They handle the **verification** of transactions, **data synchronization**, and **peer-to-peer communication**. Popular protocols like the **Bitcoin Protocol** and **Ethereum Protocol** ensure that the network remains **secure, decentralized**, and operates smoothly by allowing nodes to reach a consensus on the blockchain state.

Q52. What is Distributed Ledger Technology (DLT)?

Ans: DLT is a decentralized system for recording transactions across multiple locations.

Q53. Name any two types of Distributed Ledger Technologies.

Ans:

1. **Blockchain**
2. **Hashgraph**

Q54. How does DLT differ from traditional centralized systems?

Ans:

DLT is decentralized, meaning no single entity controls the data, whereas traditional systems rely on a central authority to manage and verify transactions.

Q55. Explain Distributed Ledger Technology and its features.

Ans:

Distributed Ledger Technology (DLT) allows data to be stored across multiple nodes or locations, without a central authority. Each participant in the network has a copy of the ledger. The key features of DLT include **decentralization**, **transparency**, **security**, and **immunity to tampering**. **Blockchain** is the most popular example of DLT, where transactions are recorded in blocks and secured using cryptography.

Q56. What is hashing in blockchain?

Ans: Hashing is the process of converting data into a fixed-size string of characters.

Q57. Name any two hashing algorithms used in blockchain.

Ans:

1. **SHA-256**
2. **RIPEMD-160**

Q58. Why is hashing important in blockchain?

Ans:

Hashing ensures data integrity, as any change in input will produce a completely different output, making tampering easy to detect.

Q59. Explain the concept of hashing in blockchain and its significance.

Ans:

Hashing in blockchain converts data (like a transaction) into a fixed-length string of characters. It is used to secure blocks of data by producing a unique hash for each block. **SHA-256** is commonly used in Bitcoin. Hashing ensures **data integrity**, as even a small change in the data results in a completely different hash, making tampering easily detectable. It also helps link blocks in a chain, ensuring the security and transparency of the blockchain.

Q60. How is blockchain used in Cryptocurrency?

Ans: Blockchain is used in Cryptocurrency to securely record and verify transactions.

Q61. Name the most famous Cryptocurrency that uses blockchain.

Ans: **Bitcoin**

Q62. Explain the role of blockchain in Cryptocurrency.

Ans:

Blockchain acts as a public ledger in Cryptocurrency, recording all transactions transparently and securely. It ensures **decentralization**, **security**, and **immutability** of transaction data.

Q63. Explain how blockchain works in Cryptocurrency.

Ans:

In Cryptocurrency, blockchain records transactions on a **decentralized ledger** across multiple computers. Each block in the chain contains transaction data, a timestamp, and a cryptographic hash of the previous block, ensuring immutability. Blockchain provides **security** through consensus mechanisms like **Proof of Work** and **Proof of Stake**, preventing fraud and double-spending. It enables secure, peer-to-peer transactions without a central authority like a bank, making crypto currencies such as **Bitcoin** and **Ethereum** possible.

Q64. What is mining in blockchain?

Ans: Mining is the process of adding new blocks to the blockchain by solving complex computational problems.

Q65. What is the purpose of mining new blocks in a blockchain?

Ans: The purpose of mining is to verify transactions, secure the network, and add new blocks to the blockchain.

Q66. What do you need to mine a new block in a blockchain?

Ans:

To mine a new block, you need a **proof of work algorithm**, a **hashing function** (like SHA-256), and a **nonce** (a random number) to find the correct block hash.

Q67. Explain the process of creating a blockchain in Python, mining new blocks, and displaying the blockchain.

Ans:

1. Creating a Blockchain:

In Python, you can create a blockchain by defining a **Block class** to represent each block with properties like `index`, `timestamp`, `data`, `previous_hash`, and `hash`.

2. Mining a New Block:

Mining involves solving a **proof of work problem**. You create a function to calculate the hash of the block and check if it meets certain conditions (e.g., leading zeros). Once the condition is met, the block is added to the chain.

3. Displaying the Blockchain:

To display the blockchain, you loop through each block in the chain and print its details like `index`, `timestamp`, `data`, `hash`, and `previous_hash`.

Python Example:

```
import hashlib
import time

class Block:
    def __init__(self, index, timestamp, data, previous_hash):
        self.index = index
        self.timestamp = timestamp
        self.data = data
        self.previous_hash = previous_hash
        self.hash = self.calculate_hash()

    def calculate_hash(self):
        return hashlib.sha256((str(self.index) + self.timestamp +
str(self.data) + self.previous_hash).encode('utf-8')).hexdigest()

class Blockchain:
    def __init__(self):
        self.chain = [self.create_genesis_block()]

    def create_genesis_block(self):
        return Block(0, time.time(), "Genesis Block", "0")

    def add_block(self, new_block):
        self.chain.append(new_block)
```



```

    def display_blockchain(self):
        for block in self.chain:
            print(f"Index: {block.index}\nTimestamp:
{block.timestamp}\nData: {block.data}\nHash: {block.hash}\nPrevious Hash:
{block.previous_hash}\n")

# Create a blockchain
my_blockchain = Blockchain()

# Add new blocks by mining
new_block = Block(1, time.time(), "Block 1 Data", my_blockchain.chain[-
1].hash)
my_blockchain.add_block(new_block)

new_block = Block(2, time.time(), "Block 2 Data", my_blockchain.chain[-
1].hash)
my_blockchain.add_block(new_block)

# Display the blockchain
my_blockchain.display_blockchain()

```

This code defines a basic blockchain, mines new blocks, and displays the whole blockchain.

Q68. How can vehicle ownership be tracked with blockchain?

Ans: Blockchain can track vehicle ownership by securely recording transactions and ownership changes in a transparent, immutable ledger.

Q69. Name one advantage of tracking vehicle ownership using blockchain.

Ans:

It provides **security** and prevents **fraud** as all ownership records are permanent and transparent.

Q70. How does vehicle ownership tracking differ with and without blockchain?

Ans:

With blockchain, ownership records are decentralized, secure, and immutable. Without blockchain, ownership tracking depends on centralized authorities, which may have security risks and can be tampered with.

Q71. Compare tracking vehicle ownership with and without blockchain.

Ans:

With Blockchain:

- Ownership is recorded on a **decentralized ledger**, ensuring **transparency, security, and immutability**.
- Transactions are **verified** by multiple parties, reducing fraud.
- It eliminates the need for intermediaries like government departments.

Without Blockchain:

- Ownership is recorded in **centralized databases** managed by authorities, which are **vulnerable to tampering** or errors.

- Data can be **lost**, leading to disputes or fraud.
- **Processing delays** can occur when verifying ownership or transferring titles.

Q72. How does blockchain help build trust?

Ans: Blockchain builds trust by providing **transparent**, **immutable**, and **secure** records of transactions.

Q73. What ensures trust in blockchain transactions?

Ans:

Trust in blockchain transactions is ensured by **cryptographic hashes**, **consensus mechanisms**, and the **decentralized nature** of the network.

Q74. Why blockchain is considered a trustworthy system?

Ans:

Blockchain is considered trustworthy because it uses **encryption**, **decentralization**, and **transparency** to prevent fraud, ensure accurate records, and allow independent verification.

Q75. Explain how blockchain works and how it builds trust.

Ans:

Blockchain works by recording transactions in **blocks**, which are linked together in a **chain**. Each block contains transaction details, a timestamp, and a cryptographic hash of the previous block. This **decentralized ledger** is accessible to all participants, making data transparent and verifiable. Blockchain builds trust by ensuring **immutability**, meaning once a transaction is recorded, it cannot be altered or erased. The use of **consensus mechanisms** like **Proof of Work** ensures that all participants agree on the data, creating an environment of **transparency** and **security**.

Q76. What is one advantage of blockchain?

Ans: Blockchain offers **security** as it uses cryptography to protect data.

Q77. Name any two advantages of blockchain.

Ans:

1. **Transparency** – All transactions are visible to participants.
2. **Decentralization** – No single entity controls the blockchain.

Q78. Mention any three disadvantages of blockchain.

Ans:

1. **Scalability issues** – Blockchain can be slow and inefficient with large transactions.
2. **High energy consumption** – Mining processes can consume a lot of energy.
3. **Regulation challenges** – Blockchain technology is still unregulated in many areas.

Q79. Explain the advantages and disadvantages of blockchain.

Ans:

Advantages:

1. **Security** – Blockchain uses encryption and hashing to secure data, making it resistant to tampering.
2. **Transparency** – All transactions are recorded in an immutable ledger visible to all participants, ensuring transparency.
3. **Decentralization** – There is no central authority, reducing single points of failure and making the system more resilient.

Disadvantages:

1. **Scalability** – Blockchain networks can face slow transaction speeds and high costs as they grow.
2. **Energy consumption** – Proof of Work algorithms in blockchain mining require significant computational power, leading to high energy usage.
3. **Regulatory challenges** – Many governments have not yet established clear regulations, which can cause uncertainty and legal challenges.

Q80. What is cybersecurity?

Ans: Cybersecurity is the practice of protecting systems, networks, and data from digital attacks.

Q81. How does blockchain contribute to cybersecurity?

Ans:

Blockchain enhances cybersecurity by providing **secure, decentralized** data storage, making it **difficult to tamper** with transaction records.

Q82. Why is cybersecurity important in blockchain?

Ans:

Cybersecurity ensures that blockchain networks remain **secure** from hacking and fraud. It protects the integrity of transactions and prevents unauthorized access to data.

Q83. Explain the role of cybersecurity in blockchain.

Ans:

Cybersecurity in blockchain ensures the **integrity, confidentiality, and availability** of data. Blockchain's **decentralized structure** and **cryptographic techniques** make it more resistant to attacks like fraud or data tampering. By using **hashing, digital signatures, and consensus mechanisms**, blockchain creates a secure environment for **peer-to-peer transactions** without a central authority. It's essential for protecting the blockchain network from cyber threats and ensuring data privacy.

Q84. How does blockchain enhance cybersecurity?

Ans: Blockchain enhances cybersecurity by providing a **decentralized and immutable** ledger for data storage, making it harder for attackers to tamper with information.

Q85. What role does encryption play in blockchain cybersecurity?

Ans:

Encryption in blockchain ensures that data is **secure** by converting it into unreadable format, protecting it from unauthorized access.

Q86. How does blockchain prevent data tampering?

Ans:

Blockchain prevents data tampering by using **cryptographic hashing** and storing each transaction in a **block** that is linked to the previous one, making alterations easily detectable.

Q87. Explain the role of blockchain in enhancing cybersecurity.

Ans:

Blockchain improves cybersecurity by providing a **decentralized network** where no single entity controls the data, reducing the risk of centralized attacks. Each transaction is secured using **cryptography** and linked in an **immutable chain**. This ensures that once data is written, it cannot be altered, providing **data integrity**. Blockchain also uses **consensus algorithms** to verify transactions, protecting the system from fraudulent activities and reducing vulnerabilities. Its transparent nature allows all participants to validate transactions, further enhancing trust and security.